LITERATURE REVIEW

A fully consistent, minimal model for non-linear market impact

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ANALYTICAL ABSTRACT

The paper asks whether a single coarse-grained model of the latent order book can reproduce, at once, the square-root law of meta-order impact, the partial decay of impact after trading stops, and the absence of price manipulation. The authors posit a reaction-diffusion dynamics for buy and sell intentions and observe that, close to the price, the net density is locally linear with slope given by a single liquidity parameter. A meta-order then enters as a source term of a heat equation, and the price satisfies one self-consistent integral equation. For constant-rate execution the impact is exactly proportional to the square root of the executed volume, with a prefactor that grows with the trading rate in the slow regime and saturates in the fast regime. Impact decay follows a propagator form at short times, while reversed trades are predicted to cost far more. The authors prove non-negative round-trip cost and separate a transient mechanical impact from a permanent informational one. This review reconstructs the derivations, verifies them numerically, and shows that the quantitative fit to observed impact requires participation rates well above those observed in practice.

1 Introduction

1.1 Paper identification and classification

The paper is a theoretical contribution to market microstructure and to the theory of price impact, at the boundary between quantitative finance and statistical physics. The field is market microstructure, the subfield is the dynamics of transient, non-linear impact of large orders split over time (meta-orders). The version read in full is the arXiv version 1412.0141 as rendered in its HTML form, including the eleven sections and the four appendices; the journal version is Donier, Bonart, Mastromatteo and Bouchaud (2015). The paper contains no new dataset: empirical facts are quoted from other studies, mostly proprietary data of Capital Fund Management. All numerical results labelled as generated in this review are the reviewer's own computations, described in the relevant tables and figure notes.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether the empirically observed square-root dependence of the price impact of a meta-order on its size, its partial relaxation after the end of execution, and its robustness to the execution rate can be derived from a single minimal dynamical model, rather than being fitted by an ad hoc non-linear modification of a linear propagator.

WHY DOES THE QUESTION MATTER?

Impact determines trading cost, capacity of strategies and the meaning of a marked-to-market price when the act of trading moves it. A model that is non-linear, time dependent and free of price manipulation is a prerequisite for sound optimal execution. The earlier theoretical benchmark, the model of Kyle (1985), gives linear and permanent impact, which contradicts the data summarised in Almgren et al. (2005) and Toth et al. (2011).

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Known: impact of meta-orders is concave, close to a square root of size, and decays afterwards (Bouchaud, Farmer and Lillo, 2009). Linear transient models, the propagator models of Bouchaud et al. (2004), have a decaying kernel but cannot generate concavity. The non-linear extension of Gatheral (2010) is ad hoc. Missing: a microscopic or mesoscopic dynamics from which the non-linearity follows and which is internally consistent with diffusive prices.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes the locally linear order book (LLOB) and a self-consistent equation for the price in the presence of a meta-order. It studies this equation analytically (small and large trading

rate, saddle-point expansion, exact scaling solution for constant rate) and shows non-negativity of round-trip cost by a sum-of-squares argument. It finds a square-root law in both regimes, a short-time decay identical to the propagator model, a stronger impact of reversed trades, and a decomposition of price moves into mechanical and informational parts.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks whether the average price trajectory y_t generated by a meta-order m_t in a market whose latent order book obeys a reaction-diffusion dynamics, in the limit of large latent liquidity and slow cancellation, (i) follows a square-root law in the executed volume, (ii) is consistent with the observed post-execution decay and reversal, and (iii) admits no profitable closed trading loop.

2.2 Hypotheses

- Local linearity: the net latent density $\varphi(y) = \rho_B - \rho_A$ is linear in the distance y to the price, with slope $-\mathcal{L}$, for a wide class of deposition and cancellation mechanisms.
- Impact is then the response of the price, defined by $\varphi(y_t, t) = 0$, to a source of orders at the price, and is non-linear because the source and the price location are coupled.
- Prices can be made diffusive by letting the common drift V_t of reservation prices be a white noise, provided a permanent informational component of impact is allowed.

2.3 Existing literature and research gap

PAPER The authors build on the zero-intelligence order book of Smith et al. (2003), reformulated as latent order book by Toth et al. (2011), on reaction-diffusion models of Bak, Paczuski and Shubik (1997), and on the agent-based derivation of Mastromatteo, Toth and Bouchaud (2014). The gap is the absence of a consistent non-linear and transient model.

CRITIQUE The gap is real for transient non-linear impact, but the paper does not compare its predictions with an existing non-linear alternative beyond the qualitative remark that Gatheral (2010) is ad hoc and yields singular optimal schedules (Curato, Gatheral and Lillo, 2017). No quantitative horse race is offered.

2.4 Claimed contribution

CRITIQUE The claimed contribution has three parts. The derivation of the square-root law in two regimes is demonstrated and reproduced numerically in this review (Table 2). The absence of price manipulation is demonstrated for the mean-field model, with a clean proof. The agreement with empirical regularities is only qualitative: no calibration is shown, and the reviewer finds (Section 8) that the participation rates at which the model reaches the empirically observed prefactor are far above realistic values.

3 Theoretical Framework and Mechanism

3.1 Latent order book dynamics

PAPER Each trading intention has a reservation price and a volume. Let $\rho_B(x,t)$ and $\rho_A(x,t)$ be the average densities of buy and sell intentions at price level x. In the hydrodynamic limit the paper writes

$$\partial_t \rho_B = -V_t\,\partial_x \rho_B + D\,\partial_x^2 \rho_B - \nu \rho_B + \lambda\,\Theta(p_t - x) - \kappa R_{AB}, \tag{1}$$

$$\partial_t \rho_A = -V_t\,\partial_x \rho_A + D\,\partial_x^2 \rho_A - \nu \rho_A + \lambda\,\Theta(x - p_t) - \kappa R_{AB}, \tag{2}$$

with the price defined by $\rho_A(p_t,t) = \rho_B(p_t,t)$.

V_t	common drift of all reservation prices (public information); a white noise makes p_t a random walk.
D	diffusion constant of reservation prices, from idiosyncratic revisions and dispersion of reactions to news.
ν	cancellation rate; $1/\nu$ is the memory of the latent book, hours to days for slow actors.
λ	deposition rate (”rain intensity”); buys arrive below the price, sells above.
κR_{AB}	reaction term, $A + B \to \emptyset$, a transaction; $R_{AB} \approx \rho_A \rho_B$ plus fluctuations, and $\kappa \to \infty$.

The drift and diffusion come from a Kramers-Moyal expansion of the master equation for agents revising $p_i \to p_i + \beta_i \xi_t + \eta_{i,t}$ (Appendix A of the paper). With a common news shock ξ_t of sensitivity β_i and idiosyncratic noise of variance Σ^2, the second-order expansion gives $\partial_t \rho = -\xi \rho' + \frac{1}{2}([\beta^2]\xi^2 + \Sigma^2)\rho''$. In the frame of the unimpacted price $\hat{p}_t = \int_0^t V_s\,ds$ only the dispersion of reactions survives,

$$D = D_0 + \frac{\sigma^2}{2} \int d\beta\,\Pi(\beta)\,(\beta - 1)^2, \tag{3}$$

so that agents who all react identically to the news ($\beta_i = 1$) do not diffuse relative to the price. The economic content is that the latent book spreads around the price only because agents disagree on how to react.

3.2 The linear combination and the stationary book

PAPER Subtracting Equation (2) from (1) removes the non-linear reaction term. The difference $\varphi = \rho_B - \rho_A$ obeys the linear equation

$$\partial_t \varphi = -V_t \, \partial_x \varphi + D \, \partial_x^2 \varphi - \nu \, \varphi + \lambda \, \mathrm{sign}(p_t - x), \qquad \varphi(p_t, t) = 0. \tag{4}$$

This is the key technical observation: although each density is governed by a non-linear equation, the price is the zero of a field obeying a linear equation with a state-dependent source. In the frame $y = x - \hat{p}_t$ the stationary solution is

$$\varphi_{\mathrm{st}}(y \le 0) = \frac{\lambda}{\nu} \left[1 - e^{\gamma y} \right], \qquad \varphi_{\mathrm{st}}(y) = -\varphi_{\mathrm{st}}(-y), \qquad \gamma^2 = \frac{\nu}{D}. \tag{5}$$

EVIDENCE Differentiating at the origin gives $|\partial_y \varphi_{\mathrm{st}}|_0 = \lambda\gamma/\nu$, hence the stationary current $J = D|\partial_y \varphi_{\mathrm{st}}|_0 = \lambda/\gamma = \lambda\sqrt{D/\nu}$ and the latent liquidity $\mathcal{L} = J/D = \lambda/\sqrt{\nu D}$. J is the volume of buy (equivalently sell) intentions executed per unit time, and $\mathcal{L}^{-1}$ plays the role of Kyle's lambda for a locally linear book. Appendix B of the paper shows that the linear behaviour at $y = 0$ survives for any deposition profile that does not diverge faster than $1/y$.

Zooming into the linear region ($\gamma \to 0$ at fixed J) gives $\varphi_{\mathrm{st}}(y) = -\mathcal{L} y$. The paper states that this is justified for meta-orders lasting up to hours and moving the price by less than a fraction of a percent, since the width of the linear region γ^{-1} is of order one percent of the price if D is of the order of volatility.

3.3 The meta-order and the central equation

PAPER A meta-order is modelled as an extra signed current m_t of orders falling exactly at the transaction price $y_t = p_t - \hat{p}_t$:

$$\partial_t \varphi(y,t) = D \, \partial_y^2 \varphi(y,t) + m_t \, \delta(y - y_t), \qquad \partial_y \varphi(y \to \pm\infty, t) = -\mathcal{L}. \tag{6}$$

The heat kernel gives the exact solution

$$\varphi(y,t) = -\mathcal{L} y + \int_0^t ds \, \frac{m_s}{\sqrt{4\pi D (t-s)}} \, \exp\left[-\frac{(y - y_s)^2}{4D(t-s)} \right], \tag{7}$$

and imposing $\varphi(y_t, t) = 0$ yields the self-consistent equation for the price,

$$y_t = \frac{1}{\mathcal{L}} \int_0^t ds \, \frac{m_s}{\sqrt{4\pi D (t-s)}} \, \exp\left[-\frac{(y_t - y_s)^2}{4D(t-s)} \right]. \tag{8}$$

EVIDENCE Equation (8) is the central result. The reading is as follows. The integrand is the Green function of the book at the current price, weighted by the trade made at time s and evaluated at the displacement $y_t - y_s$ the price has travelled since. If the displacement is small relative to the diffusion

distance, $|y_t - y_s|^2 \ll D(t-s)$, the exponential is one and the equation collapses to the linear propagator

$$y_t = \frac{1}{\mathcal{L}} \int_0^t ds \, \frac{m_s}{\sqrt{4\pi D(t-s)}}, \tag{9}$$

that is the propagator model of Bouchaud et al. (2004), with a bare kernel $G(\tau) = 1/(\mathcal{L}\sqrt{4\pi D\tau})$ decaying as $\tau^{-1/2}$. The Gaussian factor is therefore the entire non-linearity: when the price has moved far, orders placed at earlier prices contribute less, because liquidity at the new price is not what it was when they were placed. The paper also defines a volume-dependent bid and ask through $\int_{y_t^-(q)}^{y_t} \varphi \, dy = q = -\int_{y_t}^{y_t^+(q)} \varphi \, dy$, which equals $y_t \pm \sqrt{2q/\mathcal{L}}$ in equilibrium.

3.4 Constant-rate execution and the square-root law

PAPER Take $m_t = m_0 = Q/T$ on $[0, T]$. The ansatz $y_s = A\sqrt{Ds}$ solves Equation (8) exactly. Substituting $s = ut$, the displacement is $y_t - y_s = A\sqrt{Dt}\,(1-\sqrt{u})$ and the exponent becomes $A^2(1-\sqrt{u})/[4(1+\sqrt{u})]$, so

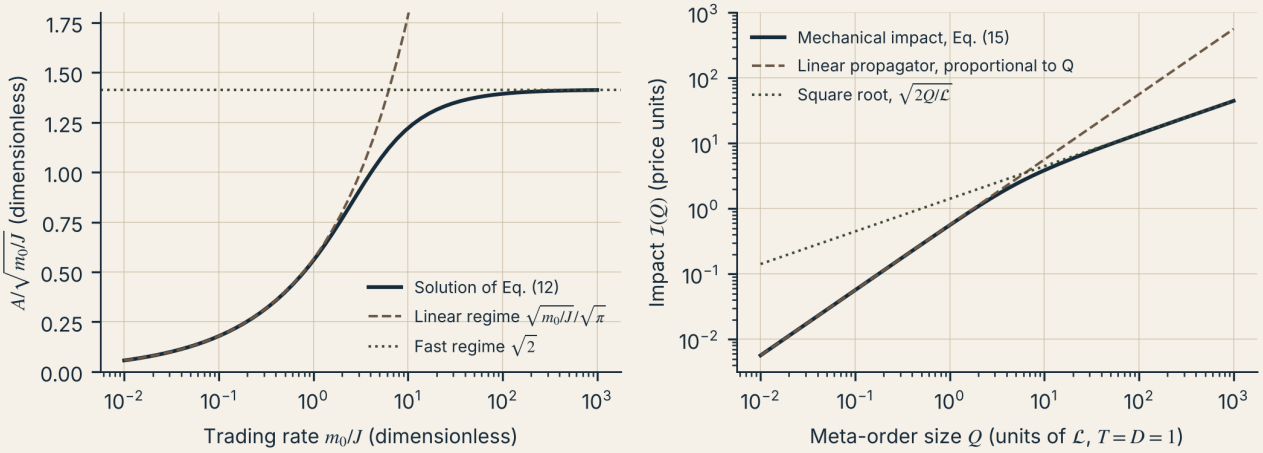
$$A = \frac{m_0}{J} \int_0^1 \frac{du}{\sqrt{4\pi(1-u)}} \exp\left[-\frac{A^2\,(1-\sqrt{u})}{4\,(1+\sqrt{u})}\right]. \tag{10}$$

EVIDENCE Two limits follow. For $m_0 \ll J$ the exponential is one and $\int_0^1 du/\sqrt{4\pi(1-u)} = 1/\sqrt{\pi}$, so $A \simeq (m_0/J)/\sqrt{\pi}$. For $m_0 \gg J$ the price is driven by the first-order balance $\mathcal{L}\, y\dot{y} = m$ of Equation (8) (see below), whose solution gives $A \simeq \sqrt{2m_0/J}$. With impact defined as $\mathcal{I}(Q) = \langle \varepsilon \cdot (p_{t+T} - p_t) \,|\, Q \rangle$ and uninformed trades, $\mathcal{I}(Q) = y_T = A\sqrt{DT} = A\sqrt{D/m_0}\,\sqrt{Q}$, hence

$$\mathcal{I}(Q) \simeq \sqrt{\frac{m_0}{\pi J}}\sqrt{\frac{Q}{\mathcal{L}}} \quad (m_0 \ll J), \qquad \mathcal{I}(Q) \simeq \sqrt{\frac{2Q}{\mathcal{L}}} \quad (m_0 \gg J). \tag{11}$$

In terms of the empirical form $\mathcal{I} = Y\sigma\sqrt{Q/V}$, with $\sigma^2 \propto D\,T_d$ and $V = J\,T_d$ the daily volume, one has $\sigma\sqrt{Q/V} = \sqrt{Q/\mathcal{L}}$, so $Y = A/\sqrt{m_0/J}$: it grows like $\sqrt{m_0/(\pi J)}$ at low rate and saturates at $\sqrt{2} \approx 1.41$ at high rate. The reviewer solved Equation (10) numerically; the result is Figure 1 and Table 2.

Figure 1. Solution of the constant-rate impact equation: the Y ratio and the impact as a function of size



Source: generated by the reviewer from Equation (12) of the paper, solved by root finding with numerical quadrature. Left: $A/\sqrt{m_0 J}$, which equals the empirical Y ratio, against m_0/J. Right: $\mathcal{I}(Q)$ for $T = D = \mathcal{L} = 1$, so $Q = m_0$ and $J = 1$. All quantities are in model units.

3.5 Large trading rates: saddle point and path independence

PAPER If $m_t \gg J$ the price displacement is large and the integral in Equation (8) is dominated by s close to t. Expanding m_s and y_s around $s = t$, the leading term is

$$m_t \int_0^\infty \frac{du}{\sqrt{4\pi D u}}\, e^{-\dot{y}_t^2 u/4D} = \frac{m_t}{|\dot{y}_t|}, \tag{12}$$

since $\int_0^\infty u^{-1/2} e^{-au} du = \sqrt{\pi/a}$ with $a = \dot{y}_t^2/4D$. Hence $\mathcal{L}\, y_t|\dot{y}_t| \approx m_t[1 + D(3\ddot{y}_t/\dot{y}_t^3 - 2\dot{m}_t/(m_t\dot{y}_t^2)) + O(J^2/m^2)]$, the paper's Equation (18). The leading term integrates to

$$y_t \simeq \sqrt{\frac{2}{\mathcal{L}} \int_0^t m_s\, ds}, \tag{13}$$

so impact depends on the total volume traded but not on the execution schedule. The cost of buying Q in time T, $\mathcal{C} = \int_0^T m_s y_s\, ds$, is then $\frac{2}{3}\sqrt{2/\mathcal{L}}\, Q^{3/2}[1 - 3J\square T/(2Q)]$ to first order, independent of the schedule (Appendix D). Intuitively, at high speed the price runs through a depleted region of the book and the cumulative volume absorbed fixes the position along a book whose profile is the linear $\varphi = -\mathcal{L}y$.

3.6 Decay and reversal

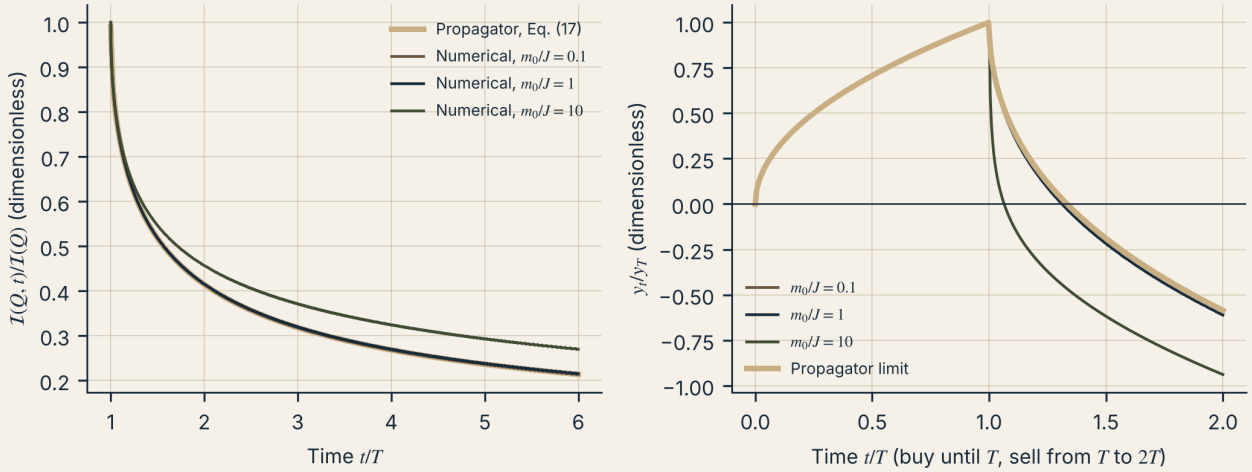
PAPER For $m_0 \ll J$ and $t > T$, Equation (9) gives $y_t = \frac{m_0}{\mathcal{L}\sqrt{4\pi D}}\, 2(\sqrt{t} - \sqrt{t - T})$, so

$$\frac{\mathcal{I}(Q, t > T)}{\mathcal{I}(Q)} = \sqrt{\frac{t}{T}} - \sqrt{\frac{t - T}{T}}, \tag{14}$$

which equals $1 - \sqrt{(t - T)/T}$ just after T (infinite initial slope) and $\frac{1}{2}\sqrt{T/t}$ at long times. Appendix C finds an exact scaling solution $\varphi(x, t) = (m_0\sqrt{t}/\sqrt{D})F(x/\sqrt{Dt})$, with $2F'' + uF' - F = -2\delta(u - A)$, and

shows that the initial decay is Equation (14) for every m_0/J, a result the authors call super-universal; the long-time prefactor depends on m_0/J. If trading is reversed at T ($m_t = -m_0$ on $[T, 2T]$) the linear model implies $y_t \propto \sqrt{t} - 2\sqrt{t-T}$, which vanishes at $t = 4T/3$, that is a delay of $T/3$ after the reversal; the paper states $T/4$ for this delay (see the critique). In the fast regime the price returns to its initial value after a delay of order $JT/(2m_0) \ll T/4$, because there is little liquidity behind the price. Round trips are therefore costly, in line with the discussion of Huberman and Stanzl (2004) and with the CFM observation reported by the authors.

Figure 2. **Decay of impact after execution and trajectory after a reversal of the meta-order**



Source: generated by the reviewer by solving Equation (9) of the paper on a time grid ($D = \mathcal{L} = 1$, $T = 1$, step 0.005, trapezoid quadrature with the singularity removed by $s = t - w^2$, root finding at each step). Left: $\mathcal{I}(Q,t)/\mathcal{I}(Q)$ against t/T. Right: y_t/y_T for buying during $[0, T]$ and selling during $[T, 2T]$. The curves for $m_0/J = 0.1$ and 1 are nearly indistinguishable from the propagator limit.

3.7 Absence of price manipulation

PAPER For a closed loop with $\int_0^T m_s\, ds = 0$ the average cost is $\mathcal{C} = \int_0^T m_s y_s\, ds$. The reviewer reconstructs the proof. Write the Gaussian as a Fourier integral, $\frac{1}{\sqrt{4\pi D\tau}} e^{-x^2/4D\tau} = \int \frac{dz}{2\pi} e^{-Dz^2\tau + izx}$, and use Equation (8):

$$\mathcal{C} = \frac{1}{\mathcal{L}} \iint ds\, du\, m_s m_u\, \Theta(s-u) \int \frac{dz}{2\pi} e^{-Dz^2(s-u)}\, e^{iz(y_s - y_u)}. \tag{15}$$

The summand is symmetric under $s \leftrightarrow u$ after taking the real part, so $\Theta(s-u)$ may be replaced by one half, with $e^{-Dz^2|s-u|}$ in the exponent. The identity $e^{-Dz^2|s-u|} = 2Dz^2 \int_{-\infty}^{\min(s,u)} dw\, e^{-Dz^2(s-w)} e^{-Dz^2(u-w)}$ turns the kernel into a sum of products, giving

$$\mathcal{C} = \frac{D}{\mathcal{L}} \int \frac{dz}{2\pi} z^2 \int dw \left| \int ds\, m_s\, \Theta(s-w)\, e^{-Dz^2(s-w) + izy_s} \right|^2 \geq 0. \tag{16}$$

This is the paper's Equations (21) and (22), with the kernel $K_z(s,u) = \Theta(s-u) e^{-Dz^2(s-u) + izy_s}$. The argument holds for the realised y_s of each schedule, so the quadratic form has a schedule-dependent kernel, but non-negativity does not require a fixed kernel. It also extends to Levy-type reservation price dynamics diagonalised in Fourier space.

3.8 Mechanical and informational impact

PAPER If the execution is correlated with the future collective drift $V_{t'}$, the apparent impact is the sum of the mechanical term and an informational term,

$$\langle p_t - p_0 \rangle = m_0 \int_0^t ds \int_0^s ds'\, C(s-s') + A(m_0)\sqrt{Dt}, \qquad C(s-s') \propto \langle V_s m_{s'} \rangle. \tag{17}$$

For $C(\tau) = \Gamma\zeta e^{-\zeta\tau}$ one obtains, for $t > T$, $\mathcal{J}_{\text{tot}} = \mathcal{J}(Q, t > T) + \Gamma Q - \frac{m_0\Gamma}{\zeta}(1 - e^{-\zeta T})e^{-\zeta(t-T)} \to \Gamma Q$. The reviewer verified this expression by integrating the correlation over $s' < T$ and $s > T$. The permanent part is linear in Q, as in Kyle (1985) and Almgren et al. (2005). The authors stress that without it a random flow of meta-orders would create a mean-reverting price component and hence market-making profits, which a permanent impact removes through adverse selection. CRITIQUE The function C is deliberately agnostic about causality; it can be alpha or a collective revaluation after trades. The model therefore cannot by itself discriminate the two, which the authors acknowledge.

4 Data and Empirical Strategy

4.1 What the paper uses

The paper has no estimation. Empirical regularities are taken from other studies: the square-root law (Torre and Ferrari, 1997; Almgren et al., 2005; Moro et al., 2009; Toth et al., 2011; Gomes and Waelbroeck, 2014; Bershova and Rakhlin, 2013; Brokmann et al., 2014), the weak dependence of the Y ratio on the trading rate in CFM data, and the Bitcoin result of Donier and Bonart (2015). Sample sizes, periods and standard errors are not reported for any of them in this paper, so none of the empirical statements can be checked from the text.

Table 1. **Model quantities, their meaning and the orders of magnitude quoted in the paper**

Quantity	Definition	Order of magnitude stated in the paper
$\mathcal{L} = J/D$	Latent liquidity, slope of the book at the price	Latent volume far above revealed volume
J	Volume executed per unit time in the stationary state	Daily volume near 1/200 of market capitalisation
$1/\nu$	Memory time of latent orders	Several hours to several days
$\gamma^{-1} = \sqrt{D/\nu}$	Width of the linear region of the book	About 1 percent of the price
m_0/J	Participation of the meta-order relative to market flow	Not given; empirical Y said to be nearly independent of it

Source: definitions reconstructed from the paper (official); the orders of magnitude are those stated by the authors in Sections III and IV, not estimates by the reviewer.

4.2 Identification and what could falsify the model

There is no identification problem in the econometric sense, because the paper makes predictions. The testable content is: (i) $\mathcal{I}(Q) \propto \sqrt{Q}$ with a prefactor given by $\sqrt{Q/\mathcal{L}}$ times a function of m_0/J bounded by $\sqrt{2}$; (ii) the normalised decay of Equation (14) at short times; (iii) the asymmetry of reversed trades. The source of variation that would identify these is exogenous variation in participation m_0/J at fixed Q, which is not available in the paper.

4.3 Limitations of the setting

- The mean-field book neglects fluctuations. The width of the book around the price vanishes as $\mathcal{L}^{-1/2}$, so the theory is valid for impacts much larger than that width.
- Look-ahead and selection do not arise in the model itself, but empirical meta-orders are chosen by traders who condition on information. The authors handle this through the informational term in Equation (17).
- Parameters D, ν, λ are treated as constants unaffected by the meta-order; the authors note that a large order may raise ν and collapse $\mathcal{L}$.

5 Results

5.1 Square-root impact

EVIDENCE The exact solution for constant-rate trading gives $\mathcal{I}(Q) = A\sqrt{D/m_0}\sqrt{Q}$. The result is statistically exact within the model; there is no sampling uncertainty. Numerically (Table 2), the impact y_T obtained by solving the full integral equation agrees with $A\sqrt{DT}$ from Equation (10) to within 0.1 percent at $m_0/J = 1$ and 10. The asymptotes are reached slowly: at $m_0/J = 10$ the ratio Y is 1.22, 14 percent below $\sqrt{2}$.

Table 2. **Reviewer verification of the main analytical claims, by numerical solution of Equation (9) with $D = \mathcal{L} = 1$**

Claim	Paper	Numerical check	Verdict
Impact at end of constant-rate execution, $m_0/J = 1$	$y_T = A\sqrt{DT}$, $A = 0.558$ from (12)	0.558 and 0.5577 from the time-stepping solver	Confirmed
Same, $m_0/J = 10$	$A = 3.862$	3.862 (quadrature) and 3.858 (solver)	Confirmed
Slow-rate limit	$A \simeq (m_0/J)/\sqrt{\pi}$	0.0564 at $m_0/J = 0.1$ against 0.0564	Confirmed
Fast-rate limit	$A \simeq \sqrt{2m_0/J}$	$A/\sqrt{m_0/J} = 1.393$ at 100 and 1.412 at 1000	Confirmed
Short-time decay (17) for all rates	Independent of m_0/J	Curves for $m_0/J = 0.1$ and 1 coincide with (17); $m_0/J = 10$ coincides initially and departs later	Confirmed
Return time of the price after reversal, linear regime	$T/4$	$0.335\,T$ at $m_0/J = 0.1$; exact algebra gives $T/3$	Not confirmed; see critique
Return time, fast regime	About $JT/(2m_0)$	$0.057\,T$ at 10 and $0.007\,T$ at 100, against $0.05\,T$ and $0.005\,T$	Consistent within grid resolution
Non-negative round-trip cost	$\mathcal{C} \geq 0$	Minimum over 25 random zero-sum schedules: $0.0073 > 0$	Confirmed
Schedule independence of cost, $Q = 200$, $T = 1$	Cost equal to first order in J/m	Constant, ramp-up, ramp-down: 2631, 2637, 2626 against leading term 2667	Confirmed to 0.4 percent

Source: generated by the reviewer (not from the paper). Solver: trapezoid quadrature in $w = \sqrt{t-s}$ with linear interpolation of the past path and Brent root finding at each time step, h between 0.0005 and 0.005. The first-order cost correction predicted is a ratio of 0.9925 at $Q = 200$; the solver gives 0.9885 at $h = 0.0025$ and 0.9905 at $h = 0.00125$, converging towards the prediction.

5.2 Economic magnitude

INFERENCE To translate the law into an intuitive quantity, take the empirical form $\mathcal{I} = Y\sigma\sqrt{Q/V}$. For a stock with daily volatility $\sigma = 2$ percent and a meta-order of 1 percent of daily volume, $\sqrt{Q/V} = 0.1$, and $Y = 1$ yields an impact of 0.2 percent, that is 20 basis points. This is an illustration with an assumed Y near the value the authors call of order unity, not a result of the paper. Under the model, however, Y is determined by m_0/J (Table 3).

Table 3. Model-implied Y ratio as a function of participation rate

Participation m_0/J	Y from Equation (10)	Linear regime $\sqrt{m_0/(\pi J)}$	Fast regime
0.01	0.056	0.056	1.414
0.1	0.178	0.178	1.414
1	0.558	0.564	1.414
3	0.901	0.977	1.414
10	1.221	1.784	1.414
100	1.393	5.642	1.414

Source: generated by the reviewer from Equation (12) of the paper by numerical solution; $Y = A/\sqrt{m_0/J}$.

5.3 Decay, reversal, and the permanent component

EVIDENCE Figure 2 shows the short-time decay following the propagator law for all rates and a long-time level that is higher when $m_0/J = 10$: at $t = 6T$ the numerical ratio is near 0.27 against 0.21 for the propagator. After reversal the price crosses its initial level at about $0.33\,T$ for $m_0/J \leq 1$ and at about $0.06\,T$ for $m_0/J = 10$, and ends at -0.94 of its peak at $2T$ for $m_0/J = 10$, against about -0.59 in the linear limit. A buy-then-sell round trip is thus considerably more expensive at high speed, which is the paper's prediction of asymmetry.

5.4 Table and figure analysis

Table 2 answers whether the paper's analytical claims survive an independent numerical solution of its own equation; the dependent variable is the price displacement y_t and the specification varies in the trading rate m_0/J and the schedule. The result that matters most is the confirmation of Equation (10) and its two asymptotes. The table cannot say whether the equation matches markets. Figure 1 (left) plots the Y ratio against the participation rate on a log axis: the relationship is a smooth crossover from a square-root growth to a plateau, the variation is entirely due to the trading rate, and it is a model relationship, not a causal estimate. The right panel shows that at fixed T the impact is linear in Q below $Q \approx JT$ and square root above it. Figure 2 (left) shows no unusual observations and the nonlinearity appears only in the long-time prefactor; (right) shows the reversal, where the non-linearity is strongest.

6 Robustness and Additional Tests

The paper offers analytical extensions rather than robustness tests in the empirical sense. They are assessed below.

Table 4. **Robustness of the central result: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Linearity of the book is an artefact of the flat deposition profile	Arbitrary deposition profiles and state dependent D, ν (Appendix B)	Linear near the price unless deposition diverges faster than $1/y$	Largely yes for the shape; no for the slope being constant in time
Dependence on the diffusion assumption	Levy-type reservation price dynamics (Appendix A)	No manipulation survives; square-root law not re-derived	Partly
Finite memory νT	Perturbation in $\nu T \ll 1$, argument for $\nu T \gg 1$ (Section X)	Impact lowered by a term proportional to νT; linear for $\nu T \gg 1$	Partly: stated, not computed
Neglect of book fluctuations	Citation of simulations in Mastromatteo et al. (2014)	Mean-field accurate for long meta-orders	Partly: relies on another paper

Concern	Test reported	Result	Does it resolve the concern?
Convexity of the book at large distance	Qualitative discussion	Would explain slower than square-root growth at large Q	No: left open
Impact of informed trades	Additive decomposition with exponential correlation (Eq. 25)	Permanent term ΓQ	Partly: functional form assumed

Source: assessment by the reviewer of the robustness discussion in Sections IV, X and Appendices A, B of the paper.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Within the mean-field LLOB, constant-rate execution gives an exactly square-root impact in both regimes, with the prefactor given by Equation (10); the short-time decay equals the propagator form for all rates; round-trip cost is non-negative for every execution schedule; and the model is a strict generalisation of the linear propagator with $\tau^{-1/2}$ kernel. These were verified by the reviewer (Table 2).

SUPPORTED BUT CONDITIONAL

That the framework explains the universal square-root law holds only if the participation rate m_0/J is of order ten or more, or if J is not the total market flow (see the main weakness). The decomposition into mechanical and permanent impact depends on an assumed correlation function.

SUGGESTIVE

The prediction that reversed trades have greater impact than the propagator model implies, and the claim that this agrees with empirical observation, rest on an internal report cited by the paper (Benichou, CFM, 2013) that cannot be examined.

NOT ESTABLISHED

The claim that the model accounts for all known empirical results is not shown, because no calibration, fit or out-of-sample test is reported. Neither is a concave-beyond-square-root regime, nor the effect of fluctuations on impact variance, nor optimal execution within the model.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- A single emergent parameter, the latent liquidity, controls the average impact, which gives a parsimonious and falsifiable model.
- The model is free of price manipulation, which makes it usable for execution problems.
- Diffusive prices coexist with order book memory through the informational component.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The reduction from a non-linear two-species problem to a linear equation for φ with a free boundary is a mathematical simplification that makes exact results possible, and the same trick is reusable in other settings.
- The Fourier proof of non-negative cost is short, rigorous within the model and extends to Levy dynamics.
- Equation (8) is easy to solve numerically; the reviewer reproduced every analytical claim in a few seconds of computation, which makes the paper unusually replicable for a theory paper.
- The model makes unusual out-of-sample predictions (reversal asymmetry, super-universal initial decay) that can be falsified.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

CRITIQUE **Problem.** In the paper's own identification $V = JT_d$ and $\sigma^2 \propto DT_d$, the Y ratio equals $A/\sqrt{m_0/J}$, which is below 0.2 for participation up to 10 percent of market flow, reaches 0.9 only at $m_0/J = 3$ and 1.22 at 10 (Table 3). Meta-orders in practice are a few percent of volume. **Why it matters.** The empirical Y is of order one and nearly rate independent; the model predicts a prefactor several times smaller and strongly rate dependent in the realistic range. The paper presents the weak rate dependence of Y as explained by the model, but this is true only in the saturated regime. **Potential bias.** Either J is not total market volume (then the identification $V = JT_d$ and the mapping of the model to the empirical Y ratio need to be revised), or the saturated regime does not describe actual meta-orders. **How it could be tested.** Estimate $\mathcal{L}$ independently, for instance from the stationary order book slope or from impact at several participation rates, and compare m_0/J and Y against Table 3.

OTHER WEAKNESSES

- Error in a stated result: the paper (as rendered in the arXiv version read) states that after a reversal the linear model returns to the initial price after $T/4$. From Equation (9) the zero of $\sqrt{t} - 2\sqrt{t-T}$ is at $t = 4T/3$, a delay of $T/3$, and the numerical solution gives $0.335\,T$. The impact of this on the conclusions is small, but it should be corrected. Test: repeat the algebra.

- No fluctuations: all results are averages of the mean-field book, while the paper itself says fluctuations are likely at the heart of liquidity crises. The mean impact may differ from the impact computed from the instantaneous price because of the book width. Test: simulate the discrete particle system.
- Persistence of parameters: $\mathcal{L}$, D, ν are fixed even though a meta-order of order JT is of the size of the latent flow and affects the behaviour of others, which would alter $\mathcal{L}$ and so the prefactor. Test: allow ν to depend on the order flow.
- Calibration absent: no parameter is estimated, so the model cannot yet be compared against the propagator alternative.
- The treatment of the permanent term is ad hoc: $C(\tau)$ is exponential by assumption and no mechanism links it to the book.

8 Contradictions and Edge Cases

CRITIQUE The square-root law disappears when $m_0 \ll J$ in the sense that the dependence on the rate becomes dominant: at fixed T the impact becomes linear in Q since $\mathcal{I} = \sqrt{m_0/(\pi J)}\sqrt{Q/\mathcal{L}}$ and $m_0 = Q/T$ give $\mathcal{I} = Q/(\sqrt{\pi}\mathcal{L}\sqrt{JT})$. The crossover is at $Q^* \approx JT$, the volume traded by the whole market during the execution time, as the right panel of Figure 1 shows. The law is therefore a statement about orders comparable to the volume of the market during the execution, not about small orders in a large market.

CRITIQUE Extreme values. For $m_0 \to \infty$ the price impact $\sqrt{2Q/\mathcal{L}}$ remains finite, so a vanishing execution time has finite impact, an artefact of neglecting that $\mathcal{L}$ itself drops (cancellation ν rises) during a fast, large order. For $Q \to 0$ impact is Q times a constant that depends on T, so the model does not reduce to a single-trade impact independent of horizon.

CRITIQUE Illustrative calculation. At $m_0/J = 0.1$ the model gives $Y = 0.178$; with $\sigma = 2$ percent and $Q/V = 1$ percent, impact is $0.178 \times 2\% \times 0.1 = 3.6$ basis points, against 20 basis points for $Y = 1$. A factor of 5.6 separates the two, larger than any sensible parameter uncertainty reported in the paper.

CRITIQUE Alternative mechanisms. Square-root impact is also generated by models of optimal execution against strategic liquidity providers and by fair-pricing arguments (Farmer et al., 2013), so the observation of the square root alone does not discriminate between the LLOB and these. The authors respond that the Bitcoin data (Donier and Bonart, 2015), where impact is below fees, makes fair-pricing arguments implausible, which is a reasonable but not conclusive argument.

CRITIQUE Outside the original setting. The model assumes continuous diffusion of reservation prices. Where order books are dominated by discrete tick constraints or where a single large agent supplies a substantial fraction of liquidity, the linear region may not exist; Appendix A discusses thin markets only briefly.

9 Further Literature

Table 5. Literature connections

Paper	Shared idea	Relationship	Why it matters
Kyle (1985)	Price impact as the solution of an information problem	Preceded; its linear permanent impact is the benchmark the paper moves away from	Gives the Kyle lambda, which here becomes $\mathcal{L}^{-1}$
Almgren and Chriss (2001)	Optimal execution with temporary and permanent impact	Preceded, applied; the model offers an alternative impact function	Defines the execution problem the model would feed
Bouchaud et al. (2004)	Propagator model with decaying kernel	Foundation; recovered as the small-impact limit	Shows the $\tau^{-1/2}$ kernel is derivable
Almgren et al. (2005)	Direct estimation of impact, square-root form	Empirical foundation	Source of the empirical law
Gatheral (2010)	No dynamic arbitrage and non-linear propagator	Competing, ad hoc non-linear generalisation	The paper argues its non-linearity is derived, not postulated
Tóth et al. (2011)	Latent order book, anomalous impact	Preceded; the same coarse-grained picture	The paper makes it dynamic and consistent
Mastromatteo, Tóth and Bouchaud (2014)	Reaction-diffusion and agent-based impact	Sibling paper, numerical support	Provides simulations of the fluctuations neglected here
Alfonsi and Schied (2010); Huberman and Stanzl (2004)	Conditions excluding price manipulation	Foundation for the no-manipulation proof	Extended to the non-linear model
Curato, Gatheral and Lillo (2017)	Optimal execution with non-linear transient impact	Related, contradicts or complements: singular optimal schedules	Motivates studying optimal execution in the LLOB
Brokmann et al. (2014); Gomes and Waelbroeck (2014)	Measured decay of impact in equities	Empirical, partly supporting	Test of the mechanical decay
Donier and Bouchaud (2015)	Bitcoin square-root law and liquidity crises	Applied, subsequent	Supports the universality claim
Bucci et al. (2019); Toth, Eisler and Bouchaud (2017)	Crossover from linear to square-root impact; universality of short-term impact	Later and related; test the crossover predicted here	Empirical test of the $Q^* \approx JT$ crossover

Source: reviewer's classification; entries from the paper's reference list are those cited by the authors (official), and the later papers are cited from the reviewer's knowledge of the literature and were not re-read for this review.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** no calibration of $\mathcal{L}$, D or ν. *Why it matters:* without it the model cannot be compared with propagator alternatives. *Existing limitation:* the authors use proprietary data and quote it only qualitatively. *Way forward:* estimate $\mathcal{L} = V/\sigma^2$ per asset (the identification implied by $V = JT_d$, $\sigma^2 = DT_d$) and test the predicted $Y(m_0/J)$.
- **Gap:** the reversal result and the super-universal decay are not tested on data. *Why it matters:* they are the most distinctive predictions. *Existing limitation:* the reversal evidence is an internal report. *Way forward:* identify round-trip meta-orders in a public dataset with account identifiers.

10.2 Literature-level gaps

- **Gap:** the relation between latent liquidity and observable liquidity measures, such as the Amihud ratio, is unclear. *Why it matters:* it would give a cross-sectional implementation. *Existing limitation:* no study links the LLOB parameters to equity characteristics. *Way forward:* cross-sectional regression with the estimator above.
- **Gap:** optimal execution inside a non-linear transient impact model that is free of manipulation. *Why it matters:* schedules in Gatheral-type models are singular. *Existing limitation:* the paper postpones it. *Way forward:* variational problem with Equation (8).

10.3 Methodological gaps

- **Gap:** fluctuations and finite $\mathcal{L}$. *Why it matters:* liquidity crises. *Existing limitation:* mean-field only. *Way forward:* Doi-Peliti or direct particle simulation of the reaction-diffusion system.
- **Gap:** numerical solution of the integral equation with singular kernel. *Why it matters:* calibration requires fast, accurate solvers. *Existing limitation:* none provided. *Way forward:* the change of variable $w = \sqrt{t-s}$ used here removes the singularity and gives a smooth integrand.

11 Research Opportunities

Idea 1. Empirical test of the participation dependence of the Y ratio

RESEARCH QUESTION. Does the Y ratio rise with $\sqrt{m_0/\pi J}$ at low participation and saturate near $\sqrt{2}$, as Equation (10) predicts, with $\mathcal{L} = V/\sigma^2$ estimated independently?

MOTIVATION. It is the cleanest direct test of the model and the reviewer finds it unlikely to hold at realistic participation.

HYPOTHESIS. Observed Y exceeds model Y at low participation by a factor above three.

ECONOMIC MECHANISM. If true, latent liquidity near the price is lower than V/σ^2 suggests, as expected if most volume is revealed high-frequency flow.

LITERATURE CONNECTION. Toth et al. (2011); Bucci et al. (2019); Donier et al. (2015).

DATA. Meta-order data with account identifiers (for example buy-side order data such as the ANcerno database, or exchanges that release member identifiers), plus daily volume and volatility.

VARIABLES. Meta-order size, duration, participation, signed price change, daily volume, volatility.

METHOD. Bin by participation; compare binned Y with Table 3; nonlinear least squares for a free scale factor on $\mathcal{L}$.

IDENTIFICATION STRATEGY. Instrument participation by exogenous variation in market volume on the day (for example index rebalancing days).

EXPECTED CONTRIBUTION. Quantitative validation or falsification of the LLOB.

MAIN RISK. Order choice is endogenous to information; the informational term contaminates the mechanical estimate.

Idea 2. Reversal and decay tests with round-trip meta-orders

RESEARCH QUESTION. Is the price return time after a reversal much shorter than the linear model implies when m_0/J is large?

MOTIVATION. Most distinctive prediction.

HYPOTHESIS. The ratio of return delay to $T/3$ falls monotonically with participation.

ECONOMIC MECHANISM. Lack of liquidity behind the price after a fast order.

LITERATURE CONNECTION. Huberman and Stanzl (2004); Brokmann et al. (2014).

DATA. Institutional trade-level data with entry and exit.

VARIABLES. Entry and exit times, sizes, signed price path.

METHOD. Event-time averaging; compare with Figure 2.

IDENTIFICATION STRATEGY. Condition on exits not triggered by price moves (for example deadline-driven liquidations).

EXPECTED CONTRIBUTION. First direct test of non-linear reversal.

MAIN RISK. Few clean round trips; exits correlate with price.

Idea 3. Optimal execution inside the locally linear order book

RESEARCH QUESTION. What is the cost-minimising schedule when impact is Equation (8) and what is the cost above the leading order?

MOTIVATION. At high speed cost is schedule independent to first order; the optimum is determined by higher-order terms.

HYPOTHESIS. The optimal schedule is regular, unlike in Gatheral-type models.

ECONOMIC MECHANISM. Slower trading lets opposing liquidity diffuse towards the price.

LITERATURE CONNECTION. Curato, Gatheral and Lillo (2017); Almgren and Chriss (2001).

DATA. None needed for the theory; calibration data for the application.

VARIABLES. $\mathcal{L}$, D, Q, T, schedule m_s.

METHOD. Variational calculus with the expansion of Appendix D and numerical optimisation of the discretised equation.

IDENTIFICATION STRATEGY. Not applicable; theoretical.

EXPECTED CONTRIBUTION. Execution algorithm grounded in a manipulation-free non-linear model.

MAIN RISK. Expansion converges slowly near $m_0 \sim J$.

Idea 4. Fluctuations of the latent book and impact variance

RESEARCH QUESTION. How large is the dispersion of impact across meta-orders of identical size, and how does it scale with $\mathcal{L}^{-1/2}$?

MOTIVATION. The mean-field theory predicts only averages, while risk depends on dispersion.

HYPOTHESIS. Impact variance scales as predicted by the reaction-diffusion front width.

ECONOMIC MECHANISM. Discreteness of orders near the price.

LITERATURE CONNECTION. Mastromatteo, Toth and Bouchaud (2014); Barkema, Howard and Cardy (1996).

DATA. Particle simulations and tick data.

VARIABLES. Impact dispersion, order book width.

METHOD. Monte Carlo of the $A + B \to \emptyset$ system with a driven source.

IDENTIFICATION STRATEGY. Compare simulated dispersion with empirical cross-sectional dispersion at matched Q/V.

EXPECTED CONTRIBUTION. Links liquidity crises to the model.

MAIN RISK. Simulation parameters unconstrained.

Idea 5. Latent liquidity as a cross-sectional pricing characteristic

RESEARCH QUESTION. Does $\mathcal{L}_i = V_i/\sigma_i^2$ contain information on expected returns or on impact cost beyond the Amihud ratio?

MOTIVATION. The model implies a specific functional form of illiquidity, $\sqrt{\sigma^2/V}$ per unit of square-rooted size, distinct from $|r|$/volume.

HYPOTHESIS. Stocks with low $\mathcal{L}$ earn a liquidity premium and have higher realised impact per unit of size.

ECONOMIC MECHANISM. Compensation for impact cost on rebalancing and for liquidity risk.

LITERATURE CONNECTION. Amihud (2002); Toth et al. (2011).

DATA. Daily stock data from CRSP, matched to execution-cost data where available.

VARIABLES. Volume, volatility, returns, size, Amihud ratio.

METHOD. Fama-MacBeth regressions and portfolio sorts on $\mathcal{L}_i$.

IDENTIFICATION STRATEGY. Control for size, turnover and Amihud; use exogenous shocks to volume such as index inclusion.

EXPECTED CONTRIBUTION. Bridges market microstructure theory and asset pricing.

MAIN RISK. $\mathcal{L}_i$ may be a re-parametrisation of known liquidity measures with no incremental content.

11.1 Ranking

Table 6. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Participation test of Y	4	4	3	4	5	5	25
2. Reversal and decay	5	3	2	4	4	4	22
3. Optimal execution	4	4	5	4	4	4	25
4. Fluctuations	4	3	4	3	3	4	21
5. Latent liquidity and returns	3	5	5	4	3	3	23

Source: reviewer's subjective scoring.

The top three are Ideas 1, 3 and 5. Idea 1 tests the model where the reviewer finds the sharpest tension and has the largest importance. Idea 3 needs no new data and builds directly on the equation and on the expansion already in the paper. Idea 5 has the best data availability and connects the model to the asset pricing literature, even if its contribution is the least certain.

12 Learning Framework

12.1 Prerequisite concepts

- Heat equation, Green functions and Fourier representation of the Gaussian.
- Reaction-diffusion systems and the $A + B \to \emptyset$ front.
- Kyle's lambda, the propagator model and the idea of a meta-order.
- Price manipulation and quasi-arbitrage (Huberman and Stanzl, 2004).
- Laplace-type and saddle-point expansions of integrals.

12.2 Core concepts learned

- The latent order book as the object that matters at low frequency, with liquidity $\mathcal{L} = J/D$.
- Impact as the response of the zero of a linear field to a source located at that zero, hence non-linearity from a linear equation.
- Two regimes, rate dependent and rate independent, of the square-root law.
- Separation of mechanical and informational components of impact.

12.3 Important equations

Equation (4) for the linear field, Equation (8) for the self-consistent price, Equation (10) for the constant-rate prefactor, Equation (11) for the two limits of impact, Equation (14) for decay and Equation (16) for the non-negative cost.

12.4 Methods learned

The change of unknown from two non-linear densities to their difference, the self-similar ansatz $y_s = A\sqrt{Ds}$, the saddle-point expansion in small J/m, the Fourier sum-of-squares proof of non-negativity, and the numerical solution of a singular Volterra-type equation after the substitution $w = \sqrt{t - s}$.

12.5 Questions to test understanding

COMPREHENSION

1. Why does the difference $\rho_B - \rho_A$ obey a linear equation although each density does not?
2. What does $\mathcal{L}$ measure, and how is it related to volume and volatility?
3. What is the difference between mechanical and informational impact?

ANALYSIS

1. Derive Equation (10) from Equation (8) with the ansatz $y_s = A\sqrt{Ds}$.
2. Show why the exponent in the Gaussian vanishes in the linear limit and why the impact then grows linearly in the rate.
3. Prove that round-trip cost is non-negative.

RESEARCH

1. What modification of the deposition profile would give impact growing more slowly than the square root?
2. How would you identify $\mathcal{L}$ from data without using meta-orders?
3. Can a single large agent create a profitable manipulation if the book is not locally linear?

12.6 Papers to read next

Toth et al. (2011); Mastromatteo, Toth and Bouchaud (2014); Curato, Gatheral and Lillo (2017); Bucci et al. (2019).

13 Conclusion

13.1 One-sentence thesis

A locally linear latent order book turns the price into the zero of a linear field driven by the meta-order, which yields a manipulation-free, non-linear and transient impact model with a square-root law.

13.2 Three most important insights

- The non-linearity is produced by a single Gaussian factor in a self-consistent equation, which collapses to the propagator model at small impact.
- Impact has two regimes, with a rate dependent prefactor $\sqrt{m_0/\pi J}$ at low participation and a plateau $\sqrt{2}$ at high participation.
- Round-trip cost is non-negative for every schedule, so the model is internally consistent.

13.3 Strongest evidence

The exact analytical solution for constant-rate trading together with the independent numerical confirmation in this review.

13.4 Biggest weakness

The saturated regime where the model reproduces an order-one Y ratio needs participation of several times market flow, far above realistic participation.

13.5 Most important research gap

Calibration of $\mathcal{L}$ and quantitative comparison with propagator models on meta-order data.

13.6 Best research opportunity

Testing the participation dependence of the Y ratio with an independent estimate of latent liquidity (Idea 1).

13.7 Three papers to read next

Toth et al. (2011); Curato, Gatheral and Lillo (2017); Bucci et al. (2019).

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